Bode Part 3: Wrapping Up the Six Building Blocks

EGR 345

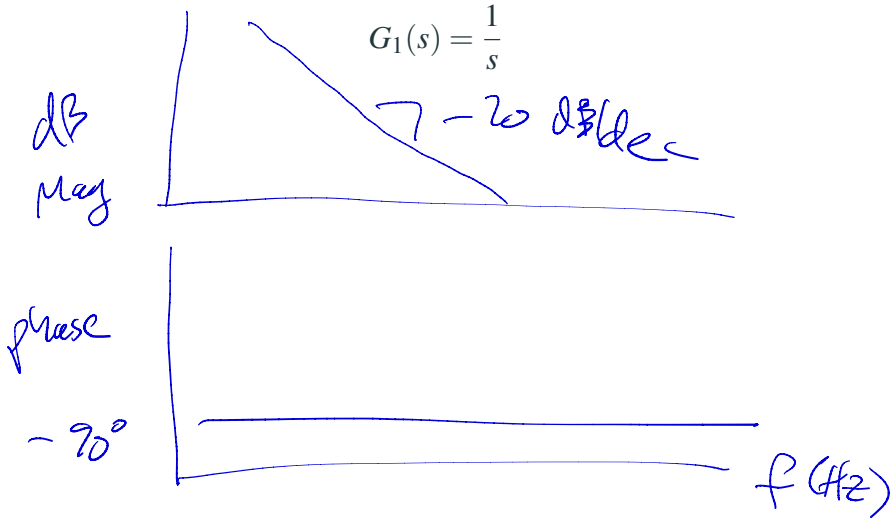
GVSU

Outline

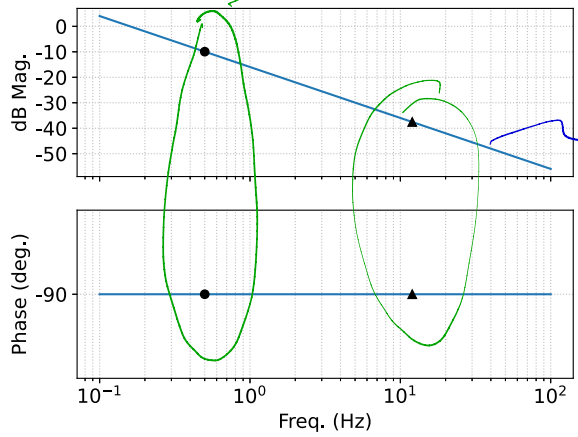
- review
- last two building blocks
- combining building blocks
- examples

Review (Slide 2)

What does this Bode plot look like?

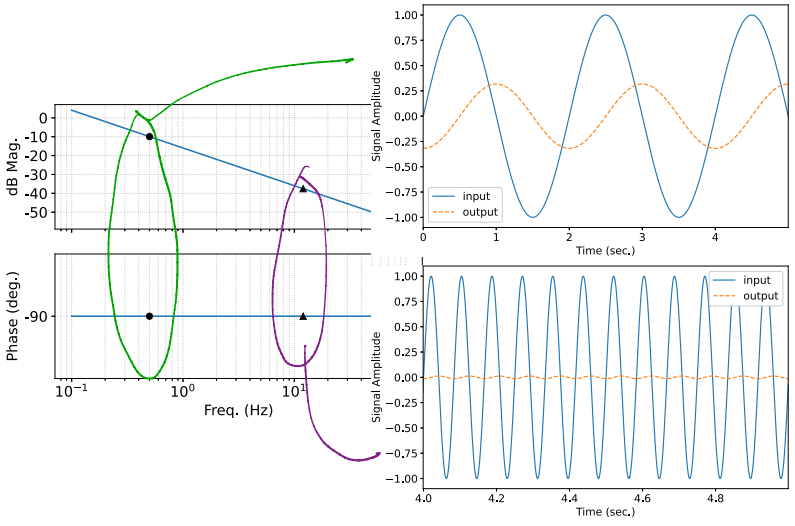


G_1 Bode

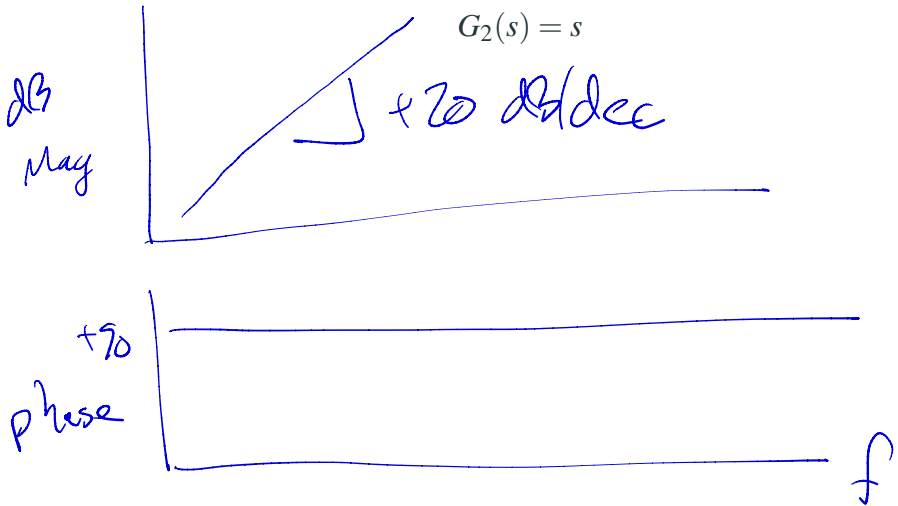


- what does the steady-state fixed sine response look like at the two marked frequencies?

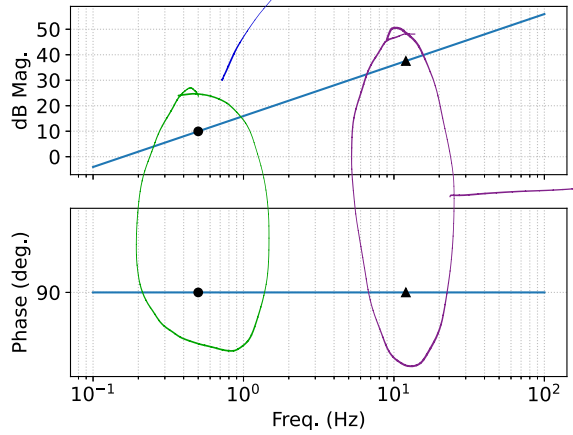
G_1 Bode and steady-state fixed sine



What does this Bode plot look like?



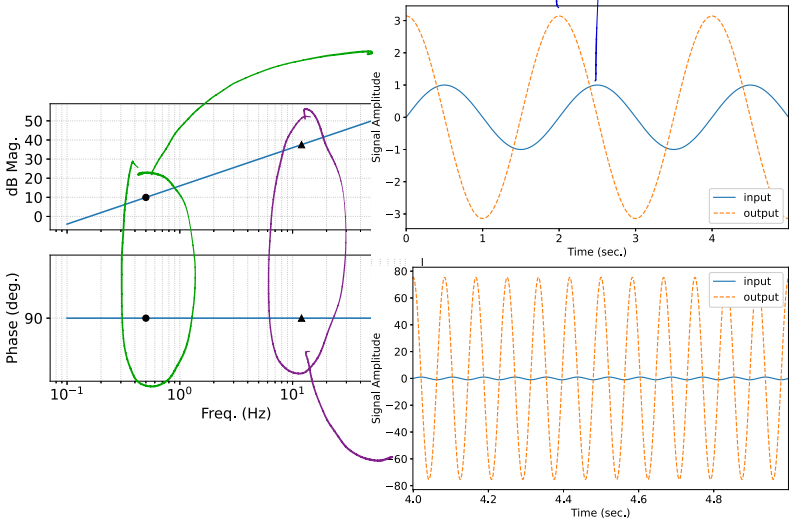
G_2 Bode



- what does the steady-state fixed sine response look like at the two marked frequencies?

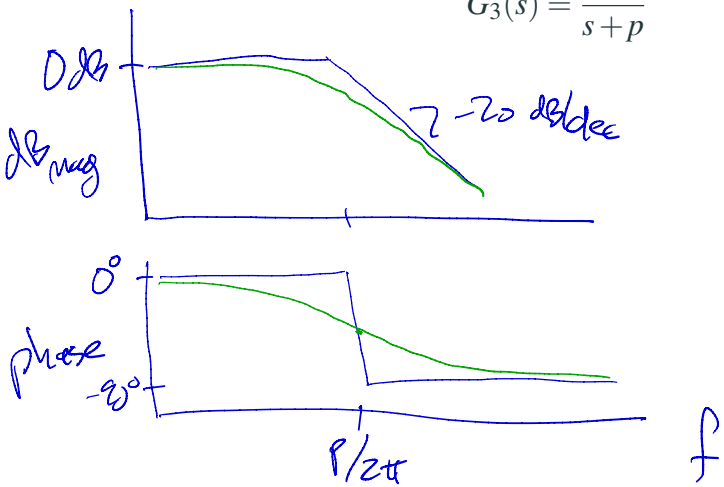
G_2 Bode and steady-state fixed sine

+90° (output leads input)

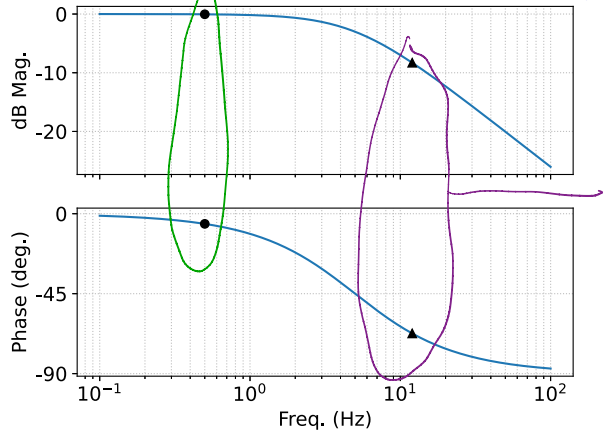


What does this Bode plot look like?

$$G_3(s) = \frac{p}{s+p}$$



G_3 **Bode**

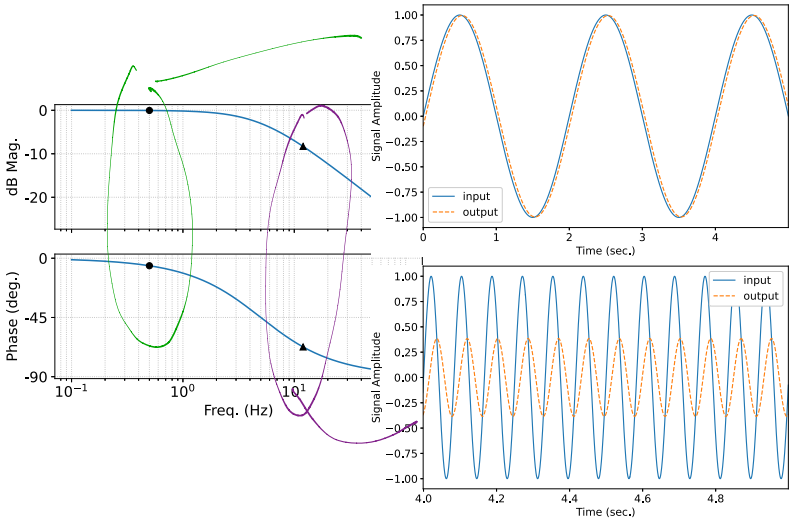


Magnitude of output ≃ input
- slight phase lag (-15°)

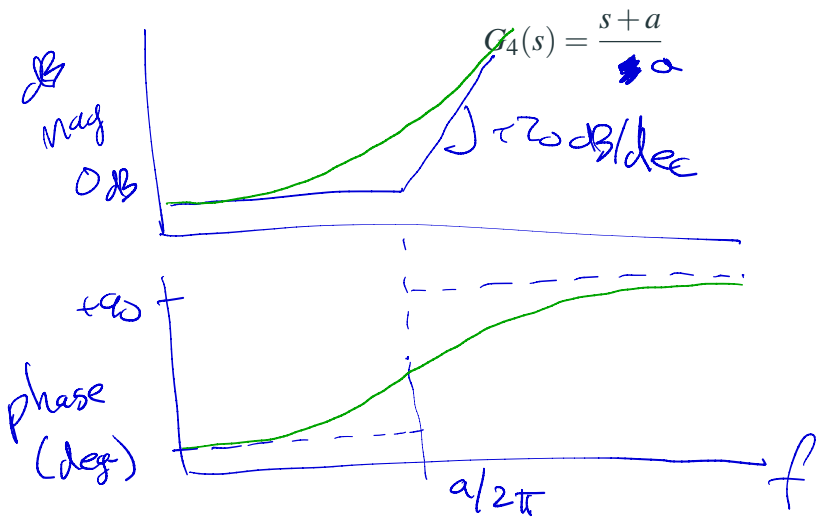
output magnitude smaller than input; ≃ -90° phase

- what does the steady-state fixed sine response look like at the two marked frequencies?

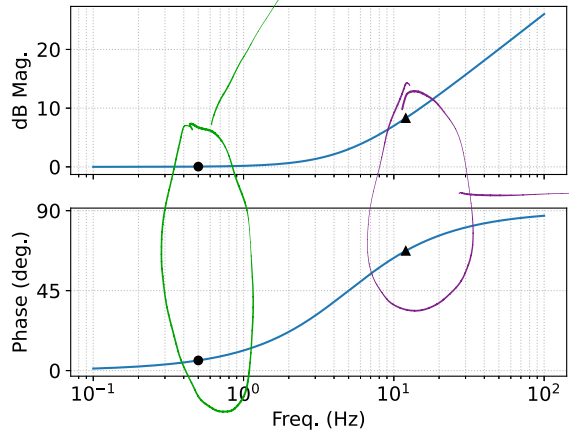
G_3 Bode and steady-state fixed sine



What does this Bode plot look like?

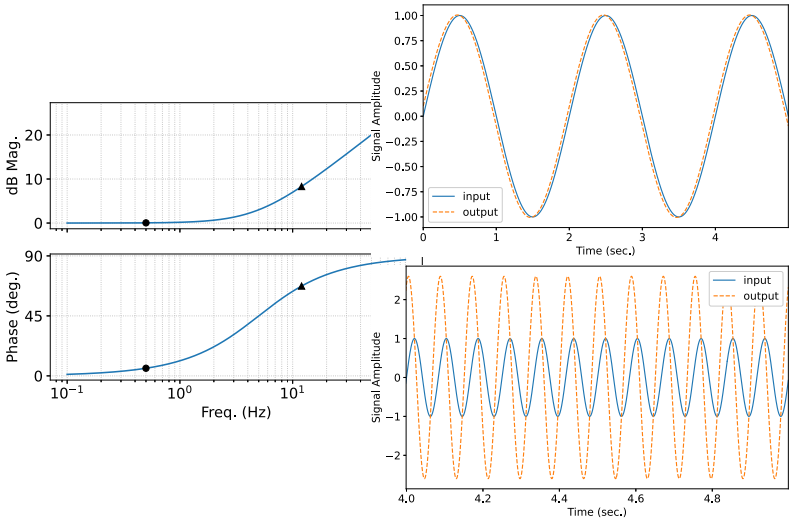


G_4 Bode



- what does the steady-state fixed sine response look like at the two marked frequencies?

G_4 Bode and steady-state fixed sine



Building Blocks 5 and 6

Building block 5: second-order (complex) poles

$$G_5(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- What happens to the magnitude and phase of $G_5(s)$ as the input frequency increases?
- three important frequencies ranges:
  - low frequency ($\omega << \omega_n$)
  - high frequency ($\omega >> \omega_n$)
  - near resonance ($\omega \approx \omega_n$)

slope = 0, phase = 0, mag = 0dB

$$G_5(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$G_5(0) = \frac{\omega_n^2}{\omega_n^2} = 1 \rightarrow 0\ dB$$

High freq

$s = j\omega,\ \omega \gg \omega_n$

$$G_5(s = j\omega,\ \omega \gg \omega_n) = \frac{\omega_n^2}{(j\omega)^2} = \frac{-\omega_n^2}{\omega^2}$$

$-40\ dB/dec$

↳ s^2 in denom. → 10x in freq. → 100x decrease in mag.

$\phi = -180°$



$$G_5(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- peak height and slope of phase transition depends on ζ



$$G_5(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Near resonance

$$s = j\omega_n$$

$$G_5(s = j\omega_n) = \frac{\omega_n^2}{-\omega_n^2 + 2\zeta\omega_n j\omega_n + \omega_n^2} = \frac{\omega_n^2}{2j\zeta\omega_n^2}$$

$$G_5(s = j\omega_n) = \frac{1}{2\zeta j} \cdot \frac{j}{j} = \frac{-j}{2\zeta}$$

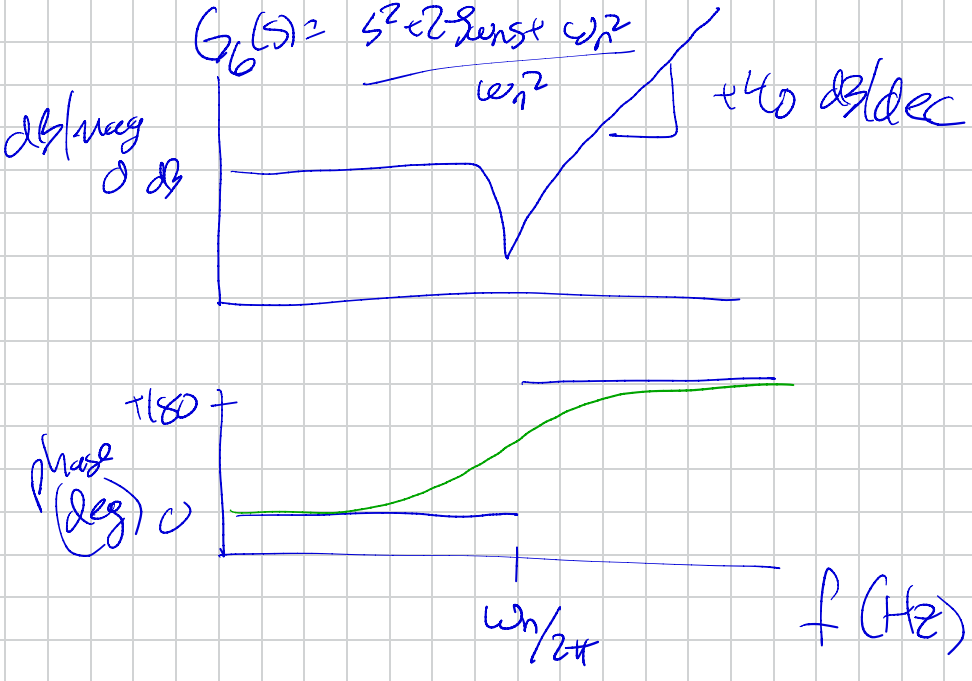
-90° phase

magnitude is inversely proportional to ζ

Building Block 6: Under-Damped Second-Order Zero

$$G_6(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{\omega_n^2}$$

- What happens to the magnitude and phase of $G_6(s)$ as the input frequency increases?
- three important frequencies ranges:
  - low frequency ($\omega << \omega_n$)
  - high frequency ($\omega >> \omega_n$)
  - near resonance ($\omega \approx \omega_n$)


$G_6(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{\omega_n^2}$
+40 dB/dec
dB/mag
0 dB
+180
phase (deg)
0
$\omega_n/2\pi$
f (Hz)

Bode Sketching Algorithm

- find all poles and zeros
  - convert them to Hz
  - sort them from lowest to highest frequency
- identify the low frequency behavior
  - pure int, pure diff. or neither
    - first or second order int. or diff. possible

initial phase and slope

- work your way up in frequency
- when you reach a pole or zero, increment or decrement the slope **and** phase
- continue until you are above the highest frequency pole or zero
- double check the high frequency slope and phase
  - slope: $-20(n-m)$ dB/dec
  - phase: $-90^\circ(n-m)$

n = # of zeros
m = # of poles

Bode Sketching Example 1

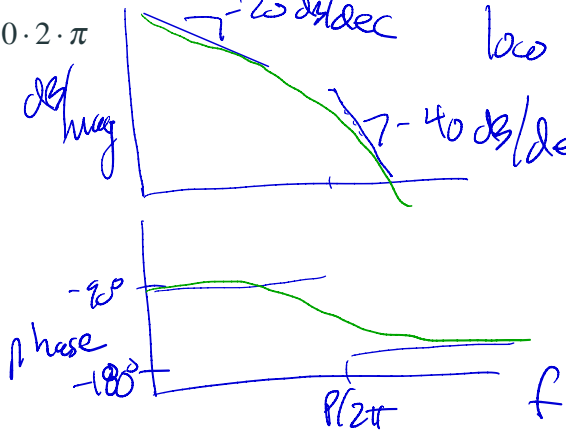
poles: $s=0$, $s=-p$

0, $p/2\pi$

zero: None ($p \neq 0$)

$$G_{ex1} = \frac{p}{s(s+p)}$$

assume $p = 10 \cdot 2 \cdot \pi$



low freq: pure int. → -20dB/dec, -90°

$n-m = 2-0 = 2$

high freq slope = -40

high freq phase = -180°

Bode Sketching Example 2

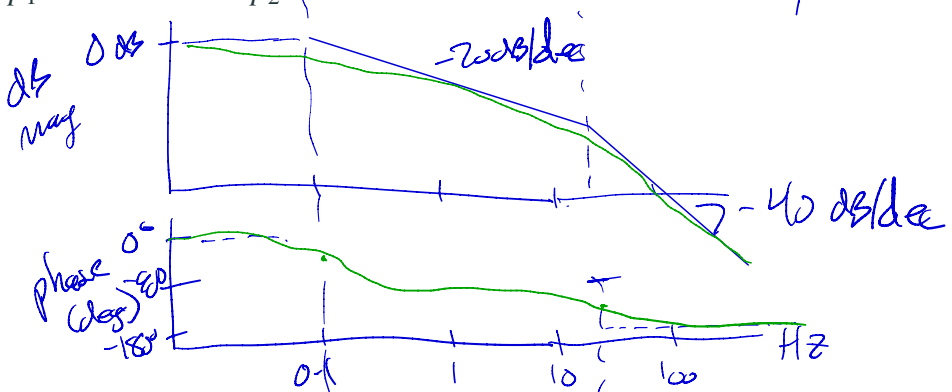
- no zeros
- poles: $-p_1$ and $-p_2$

$$G_{ex2} = \frac{p_1 \cdot p_2}{(s+p_1)(s+p_2)}$$

$p_1/2\pi$ Hz $\frac{p_2}{2\pi}$ Hz

0.1, 25

assume $p_1 = 0.1 \cdot 2 \cdot \pi$ and $p_2 = 25 \cdot 2 \cdot \pi$

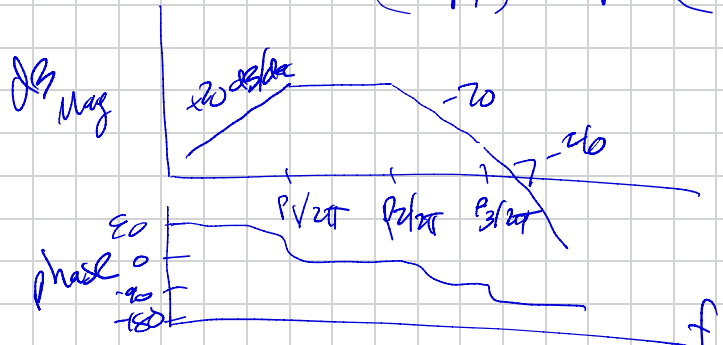


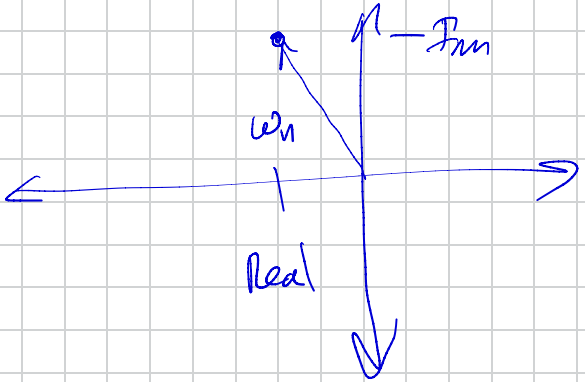
Random Bode Generator

- arguably the most instructive Python code I have ever written

Tandem Example 1

$$G = \frac{s}{(s+p_1)(s+p_2)(s+p_3)}$$





$s = -\zeta\omega_n \pm j\omega_d$

$|s| = \omega_n$

$\zeta = \frac{-\text{Real}(s)}{\omega_n}$